

Notes: Holmes (Econometrica, 2011)

The Diffusion of Wal-Mart and Economies of Density

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1 Big picture

- **Question.** Why did Wal-Mart expand through a dense, contiguous rollout rather than by jumping to distant markets, and how valuable were the **economies of density** generated by keeping stores close to distribution centers?
- **Setting.** The paper studies Wal-Mart's U.S. expansion from **1962 to 2005**, covering both regular stores and supercenters, and models the exact geography of stores and distribution centers.
- **Core challenge.** Wal-Mart's logistics cost data are not observed. So the paper cannot estimate density economies directly from an accounting relation between cost and density. It must instead infer them indirectly from observed location choices.
- **Main methodological contribution.** The paper combines
 1. a **store-level demand model** that estimates how much nearby Wal-Mart stores cannibalize one another's sales,
 2. a **dynamic location model** in which Wal-Mart chooses where to place stores given a fixed distribution-center network and fixed numbers of openings,
 3. and a **moment-inequality revealed-preference approach** that bounds the value of density economies without solving the full dynamic optimization problem.
- **Why it matters.** Store density is central to modern retail logistics. If density economies are large, policies that prevent dense rollout—for example local zoning restrictions that block entry—can impose sizable efficiency costs.
- **Main empirical question.** How much benefit does Wal-Mart obtain from having stores closer to distribution centers, once one accounts for the sales cannibalization caused by packing stores closer together?
- **Main empirical results.**
 - The baseline estimate of the distribution-cost parameter is tightly bounded at about **\$3.50–\$3.67 thousand per store-mile-year**.
 - The conservative 95% outer confidence thresholds are roughly **\$2.97–\$5.04**.
 - If all roughly **5,000 Wal-Mart store segments** were 100 miles farther from their distribution centers, annual costs would rise by almost **\$2 billion**.
 - This estimate is about **four times larger** than a pure trucking-cost calculation, implying that density economies go well beyond trucking alone.
 - A separate back-of-the-envelope calculation using distribution-center opening dates implies $\tau \approx \textbf{\$3.10}$, which closely corroborates the baseline estimate.
- **Economic takeaway.** Wal-Mart's dense, center-outward expansion pattern is consistent with **large economies of density**. Wal-Mart was willing to tolerate substantial sales cannibalization because dense store networks sharply reduced logistics costs and improved the distribution system.

2 Data and measurement (key objects)

2.1 Observed store-level, facility-level, and demographic data

- **Store-level data.** Trade Dimensions / ACNielsen provides estimated 2005 store-level sales and employment for all Wal-Mart stores in the contiguous United States.
- **Sample size.** There are **3,176 stores** in the end-of-2005 sample:
 - **1,196** regular Wal-Mart stores,
 - **1,980** supercenters.
- **Average scale.**
 - Average annual sales are **\$70.5 million** overall,
 - **\$47.0 million** for regular Wal-Marts,
 - **\$84.7 million** for supercenters.
 - Average employment is **255** workers overall,
 - **123.5** for regular stores and **333.8** for supercenters.
- **Facility opening dates.** The paper assembles opening dates for
 - regular stores,
 - food components of supercenters,
 - general distribution centers (GDCs),
 - food distribution centers (FDCs).
- **By 2005,** Wal-Mart had cumulatively opened
 - **3,176** general-merchandise store facilities,
 - **1,980** food-store facilities,
 - **43** general distribution centers,
 - **35** food distribution centers.
- **Demographics.** Census data are used at the **block-group** level for 1980, 1990, and 2000.
 - In 2000 there are **206,960** block groups.
 - Average block-group population is **1,350**.
 - Mean population density within a 5-mile radius is about **219,480 people**.
 - Mean per-capita income in 2000 dollars is about **\$21.27 thousand**.
- **Local factor prices.**
 - Wages are measured by county retail payroll divided by employment from County Business Patterns.
 - Land costs are proxied using nearby residential property values from the Census.
- **Other information.** Wal-Mart annual reports provide aggregate sales and reported cannibalization rates for comparable-store sales.

2.2 Stores, products, distribution centers, and geography

- **Two merchandise segments.**
 - g : general merchandise,
 - f : food.
- **Two store formats.**
 - A regular store sells only general merchandise.
 - A supercenter sells both general merchandise and food.
- **Distribution network.**
 - General merchandise is served by **general distribution centers**.
 - Food is served by **food distribution centers**.
- **Geographic unit.** The model uses a **continuous geography** rather than a coarse market definition such as metropolitan areas.
- **Consumer choice sets.** A consumer can shop at any Wal-Mart within **25 miles**, or choose an outside option.
- **Store rollout pattern.** Wal-Mart begins near **Bentonville, Arkansas** and expands **from the inside out**, maintaining dense local clusters and a contiguous network.

2.3 Key descriptive objects

- **Cannibalization in reported data.** Wal-Mart reported that comparable-store sales were negatively affected by new openings by about **1%** in the early 2000s.
- **Model-implied cannibalization.** The constrained demand model produces cannibalization rates of
 - **0.53** in 2001,
 - **1.02** in 2002,
 - **1.10** in 2003,
 - **1.14** in 2004,
 - **1.00** in 2005.

This fits Wal-Mart's disclosures closely.

- **Incremental versus stand-alone performance.**
 - The average new Wal-Mart generates **\$36.3 million** in incremental general-merchandise sales versus **\$41.4 million** in stand-alone sales.
 - Its incremental operating profit is **\$3.1 million** versus **\$3.6 million** stand-alone.
 - For supercenter food, the corresponding figures are **\$40.2 million** versus **\$44.8 million**, with profit **\$3.6 million** versus **\$4.0 million**.

- **Diminishing returns within states.** Stores opened later in a state's Wal-Mart history have lower incremental operating profit but are much closer to distribution centers.
- **Interpretation.** The data already show the core trade-off: denser store networks increase cannibalization and lower incremental sales, but they also keep stores much closer to the distribution system.

2.4 Unobservables and timing

- **Unobserved demand component.** Store-level sales data contain classical measurement error because Trade Dimensions sales are estimates rather than exact scanner totals.
- **Unobserved cost component.** County wages and land-rent proxies are measured with error at the store level.
- **Timing and dynamics.**
 - Wal-Mart chooses store openings over time.
 - The analysis conditions on the **actual number of openings** and the **actual distribution-center opening dates**.
 - Once a store is opened, it is assumed not to close.
 - There is an exogenous growth factor ρ_t capturing expansion in Wal-Mart's business model over time.
- **Pricing assumption.** Wal-Mart prices are treated as constant across stores and over time, motivated by its everyday-low-price policy.

3 Models and empirical specifications (all equations + interpretation)

3.1 Demand system

A consumer at location ℓ compares the outside option with all Wal-Mart stores within 25 miles. Outside-option utility is

$$u_0 = b(\text{Popden}_\ell) + \text{LocationChar}_\ell \alpha + \varepsilon_0,$$

with

$$b(\text{Popden}) = \alpha_0 + \alpha_1 \ln(\text{Popden}) + \alpha_2 [\ln(\text{Popden})]^2.$$

Utility from shopping at Wal-Mart store j is

$$u_{\ell j} = -h(\text{Popden}_\ell) \text{Distance}_{\ell j} + \text{StoreChar}_j \gamma + \varepsilon_j,$$

where

$$h(\text{Popden}) = \xi_0 + \xi_1 \ln(\text{Popden}).$$

Predicted general-merchandise revenue for store j is

$$R_j^g = \sum_{\{\ell | j \in \bar{B}_\ell^{\text{Wal}}\}} \lambda^g p_{j\ell}^g n_\ell,$$

and food revenue is defined analogously for supercenters.

Interpretation.

- Demand falls with distance to a Wal-Mart.
- The outside option becomes more attractive in denser areas, capturing the fact that urban consumers have more substitutes.
- Nearby Wal-Mart stores are substitutes, so packing stores together lowers each store's sales through cannibalization.

3.2 Distribution costs, variable costs, and fixed costs

For a supercenter, distribution cost is

$$\text{DistributionCost}_{jt} = \tau d_{jt}^g + \tau d_{jt}^f,$$

and for a regular store it is simply τd_{jt}^g .

Variable store inputs are proportional to sales:

$$\text{Labor}_j = \nu^{\text{Labor}} R_j, \quad \text{Land}_j = \nu^{\text{Land}} R_j, \quad \text{Other}_j = \nu^{\text{Other}} R_j.$$

Fixed cost varies with local population density as

$$c(\text{Popden}_j) = \omega_0 + \omega_1 \ln(\text{Popden}_j) + \omega_2 [\ln(\text{Popden}_j)]^2.$$

Interpretation.

- τ is the key density-economy parameter: it measures the annual benefit from a store being 1 mile closer to its nearest distribution center.
- ω_1 and ω_2 capture the idea that very dense urban areas may be intrinsically disadvantageous for Wal-Mart, even after accounting for wages and land prices.
- Distribution costs are modeled as fixed with respect to store sales volume, reflecting the importance of daily delivery runs.

3.3 Dynamic location problem

Let a denote a feasible policy vector listing Wal-Mart's store openings over time. Wal-Mart solves

$$\max_a \sum_{t=1}^{\infty} (\rho_t \beta)^{t-1} \left[\sum_{j \in B_t^{\text{Wal}}} \left(\pi_{jt}^g - c_{jt}^g - \tau d_{jt}^g \right) + \sum_{j \in B_t^{\text{Super}}} \left(\pi_{jt}^f - c_{jt}^f - \tau d_{jt}^f \right) \right],$$

where operating profit for merchandise segment $e \in \{g, f\}$ is

$$\pi_{jt}^e = \mu R_{jt}^e - \text{Wage}_{jt} \text{Labor}_{jt}^e - \text{Rent}_{jt} \text{Land}_{jt}^e - \text{Other}_{jt}^e.$$

Interpretation.

- The paper focuses on **where** Wal-Mart opens stores, not on **how many** stores it opens or **when** it opens distribution centers.
- Those other choices are conditioned on their observed values.
- The density-economy parameter enters only through the distance between stores and their nearest distribution centers.

3.4 Demand estimation and cannibalization

The demand parameters are estimated by maximum likelihood using 2005 store-level sales. Because store sales in the data are estimates, the discrepancy between model-predicted and observed log sales is treated as classical measurement error.

The paper defines the model's **cannibalization rate** for a year as the percentage reduction in comparable-store sales at preexisting stores caused by new openings and supercenter conversions in that year.

Interpretation.

- Cannibalization is the key observed **cost** of high density.
- If Wal-Mart tolerates large cannibalization, then the unobserved **benefit** of density must also be large.
- The constrained demand model forces the 2005 cannibalization rate to equal 1 and is used as the conservative baseline.

3.5 Incremental sales and incremental operating profits

For any store opening, the paper computes the increment to total Wal-Mart sales and operating profits relative to a counterfactual in which that store were not opened in that year.

Interpretation.

- Incremental sales are smaller than stand-alone sales because some demand is diverted from existing stores.

- Comparing incremental operating profit to incremental distribution distance gives a first look at the trade-off Wal-Mart faced when choosing denser versus sparser rollout.

3.6 Moment-inequality framework

For a generic deviation a from Wal-Mart's chosen policy a° , revealed preference implies

$$y_a \geq x'_a \theta,$$

where y_a is the profit gain from the chosen policy over the deviation, x_a is the corresponding change in cost-relevant objects, and

$$\theta = (\tau, \omega_1, \omega_2).$$

With nonnegative instruments z_{ak} , the population moment inequalities are

$$m_k(\theta) = E[z_{ak}\tilde{y}_a] - E[z_{ak}x'_a]\theta \geq 0.$$

The identified set is the set of θ satisfying all moment inequalities. Sample analog estimation uses

$$\tilde{Q}(\theta) = \sum_{k=1}^K (\min\{0, \tilde{m}_k(\theta)\})^2.$$

Interpretation.

- The model does not point-identify τ from one exact first-order condition.
- Instead, it uses many revealed-preference inequalities from deviations in Wal-Mart's rollout.
- Moment inequalities are essential because store-level wages, rents, and first-stage demand estimates all contain measurement error.

3.7 Mapping the Wal-Mart problem into the inequalities

The incremental present value of operating profit is

$$y_a = \Pi(a^\circ) - \Pi(a),$$

with

$$\Pi(a) = \sum_{t=1}^{\infty} (\rho_t \beta)^{t-1} \left(\sum_{j \in B_t^{\text{Wal}}(a)} \pi_{jt}^g(a) + \sum_{j \in B_t^{\text{Super}}(a)} \pi_{jt}^f(a) \right).$$

The cost vector is

$$x_{1a} = \Delta D_a, \quad x_{2a} = \Delta C_{1a}, \quad x_{3a} = \Delta C_{2a},$$

where

- ΔD_a is the present-value difference in distribution miles,
- ΔC_{1a} is the present-value difference in $\ln(\text{Popden})$,
- ΔC_{2a} is the present-value difference in $[\ln(\text{Popden})]^2$.

Interpretation.

- Each deviation changes the sequence of opening dates of a pair of stores.
- If Wal-Mart chose the observed sequence rather than the deviation, the observed sequence must weakly dominate in expected objective value.
- This directly turns store-opening behavior into information about τ , ω_1 , and ω_2 .

3.8 Deviation design and instruments

The paper uses **pairwise resequencing deviations**:

- a store that actually opened earlier is delayed,
- another store is moved earlier,
- all other openings are left unchanged.

There are **12 basic deviation groups**:

- store-density-decreasing groups,
- store-density-increasing groups,
- population-density-changing groups.

Basic instruments are group indicators multiplied by weights. Additional **level 1** and **level 2** interactions multiply these group indicators by positive transformations of the cost variables.

Interpretation.

- The store-density deviations isolate the trade-off between operating-profit losses and distance savings.
- The population-density deviations help separately identify the urban fixed-cost parameters ω_1 and ω_2 .
- Interaction moments dramatically tighten the identified set.

3.9 Distribution-center opening cross-check

Holding store openings fixed, opening a distribution center k in year t_k^{open} implies approximately

$$\phi_k \approx \tau D_{k,t}^{\text{inc}},$$

where $D_{k,t}^{\text{inc}}$ is the incremental reduction in store miles from opening that distribution center.

Interpretation.

- This is a separate revealed-preference condition based on distribution-center timing rather than store timing.
- It provides a useful back-of-the-envelope corroboration for the main estimate of τ .

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- **No direct logistics-cost data.** Wal-Mart does not disclose confidential distribution costs, so one cannot simply regress logistics cost on density.
- **Huge dynamic choice space.** The number of possible store-opening sequences is enormous. Directly solving Wal-Mart's dynamic problem is computationally infeasible.
- **Exact revealed preference with noisy data fails.** If one imposed every disaggregated inequality store by store, outlier measurement error would dominate and the estimates would break down.

4.2 What identifies the new approach

- **Demand estimation identifies cannibalization.** The first stage recovers how much sales are lost when stores are placed near one another.
- **Observed rollout identifies trade-offs.** Pairwise resequencing deviations compare the operating-profit consequences of denser versus less dense rollout patterns.
- **Urban-density shifts identify fixed costs.** Deviations within states that keep store density roughly constant but change population density separately pin down ω_1 and ω_2 .
- **Moment inequalities average out measurement error.** Nonnegative instruments and zero-mean measurement error make the aggregated inequalities valid.

4.3 Why partial identification is useful here

- The approach does not require point identification from a fully solved dynamic model.
- Instead, it produces a set of parameter values consistent with Wal-Mart's observed choices.
- With only 12 basic moments the identified set is fairly wide, but once level 1 and level 2 interaction moments are added, the bound on τ becomes extremely tight.

4.4 Empirical credibility of the application

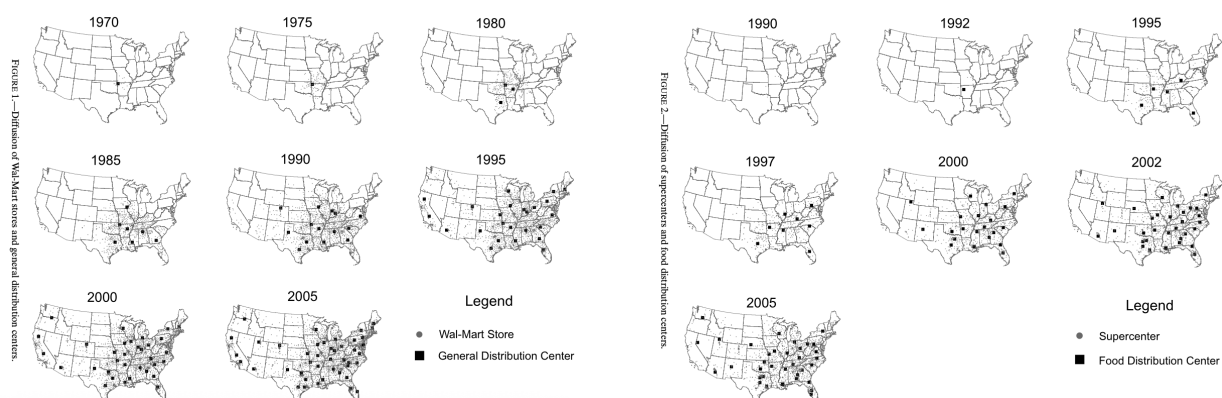
- **Demand model fit.** The constrained demand model fits Wal-Mart's reported cannibalization rates closely.
- **Subsampling for dependence.** Many deviations share stores, so moment conditions are not independent. The paper uses a subsampling procedure over stores rather than deviations to estimate variance correctly.
- **Stage 1 and stage 2 error correction.** Inference incorporates both first-stage demand-estimation error and second-stage wage/rent measurement error.
- **Cross-check from DC openings.** The alternative calculation using distribution-center openings delivers a very similar implied τ , supporting the baseline.

4.5 Relation to the literature

- The paper uses the partial-identification and moment-inequality approach of **Pakes, Porter, Ho, and Ishii (2006)**.
- It is closely related to **Jia (2008)**, but differs by emphasizing **dynamics** and **cannibalization** rather than static location competition.
- It contributes to the literature on **economies of density** by explicitly modeling the distribution-system channel in a very fine geographic environment.

5 Tables and figures

5.1 Figure 1 and Figure 2: diffusion of stores, supercenters, and distribution centers



Read:

- Figure 1 shows the expansion of regular Wal-Mart stores and general distribution centers. The key visual fact is **inside-out diffusion**: Wal-Mart expands outward from Arkansas while maintaining a dense contiguous network.

- Figure 2 shows the same pattern for the supercenter format and food distribution centers beginning in 1988.
- The main punchline is that the observed rollout is not random geographic spread; it is systematically organized around density and contiguity.

5.2 Table I, Table II, and Table III: store scale, facility growth, and demographics

TABLE I
SUMMARY STATISTICS OF STORE-LEVEL DATA^a

Store Type	Variable	N	Mean	Std. Dev.	Min	Max
All	Sales (\$millions/year)	3,176	70.5	30.0	9.1	166.4
Regular Wal-Mart	Sales (\$millions/year)	1,196	47.0	20.0	9.1	133.9
Supercenter	Sales (\$millions/year)	1,980	84.7	25.9	20.8	166.4
All	Employment	3,176	254.9	127.3	31.0	812.0
Regular Wal-Mart	Employment	1,196	123.5	40.1	57.0	410.0
Supercenter	Employment	1,980	333.8	91.5	31.0	812.0

^aEnd of 2005, excludes Alaska and Hawaii. Source: Trade Dimensions retail data base.

TABLE II
DISTRIBUTION OF WAL-MART FACILITY OPENINGS BY DECADE AND OPENING TYPE^a

Decade Open	General Merchandise (Including Supercenters)		Food Store (Part of Supercenter)		General Distribution Centers		Food Distribution Centers	
	Opened This Decade		Opened This Decade		Opened This Decade		Opened This Decade	
	Decade	Cumulative	Decade	Cumulative	Decade	Cumulative	Decade	Cumulative
1962–1969	15	15	0	0	1	1	0	0
1970–1979	243	258	0	0	1	2	0	0
1980–1989	1,082	1,340	4	4	8	10	0	0
1990–1999	1,130	2,470	679	683	18	28	9	9
2000–2005	706	3,176	1,297	1,980	15	43	26	35

^aSource: See Appendix A.

Read:

- Table I establishes the scale of Wal-Mart's operation by 2005: **3,176 stores**, high average sales, and large differences between regular stores and supercenters.
- Table II shows the time structure of expansion: regular-store openings surge in the 1980s and 1990s, while food-store and food-distribution openings come later with the supercenter rollout.
- Table III provides the consumer-location data used in the demand model and shows that population density varies enormously across block groups.

5.3 Table IV and Table V: demand estimates and cannibalization

Read:

- Table IV shows that the demand model fits store-level sales well. In the constrained specification, estimated spending per consumer is about $\lambda^g = 1.938$ and $\lambda^f = 1.912$ thousand dollars annually, the distance-disutility constant is $\xi_0 = 0.703$, and the fit is strong with $R^2 \approx 0.753$.
- The outside option becomes more attractive in denser places, which is exactly what the model needs to rationalize weaker Wal-Mart demand in urban areas.

TABLE III
SUMMARY STATISTICS FOR CENSUS BLOCK GROUPS^a

	1980	1990	2000
N	269,738	222,764	206,960
Mean population (1,000)	.83	1.11	1.35
Mean population density (1,000 in 5-mile radius)	165.3	198.44	219.48
Mean per capita income (thousands of 2000 dollars)	14.73	18.56	21.27
Share old (65 and up)	.12	.14	.13
Share young (21 and below)	.35	.31	.31
Share black	.10	.13	.13

^aSource: "Summary File 3, Census of Population and Housing" (U.S. Bureau of the Census (1980, 1990, 2000)).

^bPer capita income is truncated from below at \$5,000 in year 2000 dollars.

TABLE IV
PARAMETER ESTIMATES FOR DEMAND MODEL

Parameter	Definition	Unconstrained	Constrained (Fits Reported Cannibalization)
λ^g	General merchandise spending per person (annual in \$1,000)	1.686 (.056)	1.938 (.043)
λ^f	Food spending per person (annual in \$1,000)	1.649 (.061)	1.912 (.050)
ξ_0	Distance disutility (constant term)	.642 (.036)	.703 (.039)
ξ_1	Distance disutility (coefficient on $\ln(\text{Popden})$)	-.046 (.007)	-.056 (.008)
α	Outside alternative valuation parameters		
	Constant	-8.271 (.508)	-7.834 (.530)
	$\ln(\text{Popden})$	1.968 (.138)	1.861 (.144)
	$\ln(\text{Popden})^2$	-.070 (.012)	-.059 (.013)
	Per capita income	.015 (.003)	.013 (.003)
	Share of block group black	.341 (.082)	.297 (.076)
	Share of block group young	1.105 (.464)	1.132 (.440)
	Share of block group old	.563 (.380)	.465 (.359)
γ	Store-specific parameters		
	Store age 2 + dummy	.183 (.024)	.207 (.023)
σ^2	Measurement error	.065 (.002)	.065 (.002)
N		3,176	3,176
Sum of squared error		205.117	206.845
R^2		.755	.753
(Likelihood)		-155.749	-169.072

TABLE V
CANNIBALIZATION RATES, FROM ANNUAL REPORTS AND IN MODEL^a

Year	From Annual Reports	Demand Model (Unconstrained)	Demand Model (Constrained)
1998	n.a.	.62	.48
1999	n.a.	.87	.67
2000	n.a.	.55	.40
2001	1	.67	.53
2002	1	1.28	1.02
2003	1	1.38	1.10
2004	1	1.43	1.14
2005	1	1.27	1.00 ^b

^aSource: Estimates from the model and Wal-Mart Stores, Inc. (1971–2006) (Annual Reports 2004, 2006).

^bCannibalization rate is imposed to equal 1.00 in 2005.

- Table V is especially important: the estimated cannibalization rates line up closely with Wal-Mart’s own reported “approximately 1%” figure for the early 2000s.
- This gives credibility to the first-stage demand model and justifies using the constrained demand model as the baseline.

5.4 Table VI and Table VII: demand comparative statics and variable costs

TABLE VI
COMPARATIVE STATICS WITH DEMAND MODEL^a

Distance (Miles)	Population Density (Thousands of People Within a 5-Mile Radius)						
	1	5	10	20	50	100	250
0	.999	.989	.966	.906	.717	.496	.236
1	.999	.979	.941	.849	.610	.387	.172
2	.997	.962	.899	.767	.490	.288	.123
3	.995	.933	.834	.659	.372	.206	.086
4	.989	.883	.739	.531	.268	.142	.060
5	.978	.803	.615	.398	.184	.096	.041
10	.570	.160	.083	.044	.020	.011	.006

^aUses constrained model.

TABLE VII
DISTRIBUTION OF VARIABLE INPUT COSTS^a

Estimated 2005 Labor Costs			
Quartile	Store Location	Annual Payroll per Worker (\$)	Wages as Percentage of Sales
Minimum	Pineville, MO	12,400	4.5
25	Litchfield, IL	19,300	7.0
50	Belleville, IL	21,000	7.6
75	Miami, FL	23,000	8.3
Maximum	San Jose, CA	37,900	13.7
Estimated 2005 Land Value–Sales Ratios			
Quartile	Store Location	Index of Residential Property Value per Acre (\$)	Land Value as Percentage of Sales
Minimum	Lincoln, ME	1,100	.0
25	Campbellsville, KY	32,100	1.2
50	Cleburne, TX	67,100	2.4
75	Albany, NY	137,300	5.0
Maximum	Mountain View, CA	1,800,000	65.0

^aPercentiles of distribution are weighted by sales revenue.

TABLE VIII
INCREMENTAL AND STAND-ALONE VALUES OF NEW STORE OPENINGS^a

	<i>N</i>	Incremental Sales	Operating Profit	Incremental Distribution Center Distance (Miles)	Stand-Alone Sales	Stand-Alone Operating Profit
Part A: General Merchandise (New Wal-Marts Including Supercenters)						
All	3,176	36.3	3.1	168.9	41.4	3.6
By state's Wal-Mart age at opening						
1-2	288	38.0	3.5	343.3	38.7	3.6
3-5	614	39.5	3.5	202.0	41.5	3.7
6-10	939	37.6	3.3	160.7	40.9	3.6
11-15	642	36.1	2.9	142.1	42.2	3.4
16-20	383	32.9	2.8	113.7	41.2	3.5
21 and above	310	29.5	2.4	90.2	44.4	3.6
Part B: Food (New Supercenters)						
All	1,980	40.2	3.6	137.0	44.8	4.0
By state's Wal-Mart age at opening						
1-2	202	42.4	3.9	252.9	43.9	3.9
3-5	484	42.7	4.0	171.2	44.7	4.1
6-10	775	41.0	3.6	113.5	45.3	4.0
11-15	452	36.7	3.2	95.3	45.3	3.9
16-20	67	30.1	2.8	94.0	38.6	3.5

^aAll evaluated at 2005 demand equivalents in millions of 2005 dollars.

Read:

- Table VI shows how sharply demand varies with population density and distance.
 - In a very rural area, a consumer located at the store has shopping probability essentially **1**.
 - At a density of **250,000** people within 5 miles, that probability falls to only **0.236** even when distance is zero.
 - At that same high density, increasing distance from 0 to 5 miles drops predicted demand to about **0.041**.
- Table VII shows that median labor cost is about **7.6% of sales**, while median land value is about **2.4% of annual sales**. Since the paper assumes annual land rent is 20% of land value, land rent is a relatively small part of operating cost for the median store.
- The broader message is that both local demand conditions and local input costs vary sharply across space, so Wal-Mart's rollout problem is genuinely geographic.

5.5 Table VIII: preliminary trade-off between cannibalization and density

Read:

- The average new Wal-Mart has incremental general-merchandise sales of **\$36.3 million** versus stand-alone sales of **\$41.4 million**; incremental operating profit is **\$3.1 million** versus stand-alone **\$3.6 million**.
- For supercenter food, incremental sales are **\$40.2 million** versus stand-alone **\$44.8 million**, and incremental operating profit is **\$3.6 million** versus **\$4.0 million**.
- By state age, incremental operating profit declines as Wal-Mart opens later stores in a state, but distance to the nearest distribution center also declines substantially.

- The paper interprets this as clear preliminary evidence of a trade-off: later stores are more cannibalized, but they are much cheaper to serve through the distribution network.

5.6 Table IX and Table X: deviation design and moment construction

TABLE IX
DEFINITIONS OF DEVIATION GROUPS^a

Deviation Category	Deviation Group	Description (Store j' Flips with Store j)
Store density decreasing		Find the set of stores, $S = \{j, t_j \geq t_j^{\text{state}} + 10\}$. For each $j \in S$, find all j' , where (i) $t_{j'} \geq t_j + 3$, (ii) j' is in a different state than j , and (iii) $t_{j'} \leq t_j^{\text{state}} + 4$. Take all of these and further classify by group on the basis of ΔD_s as follows:
	1	$-.75 \leq \Delta D_s < 0$
	2	$-1.50 \leq \Delta D_s < -.75$
	3	$\Delta D_s < -1.50$
Store density increasing		Find the set of stores, $S = \{j, t_j \leq t_j^{\text{state}} + 5\}$. For each $j \in S$, find all j' , where (i) $t_{j'} \geq t_j + 3$, (ii) j' is in a different state than j , and (iii) $t_{j'} \geq t_j^{\text{state}} + 10$. Take all of these and further classify by group on the basis of ΔD_s as follows:
	4	$0 < \Delta D_s \leq .75$
	5	$.75 < \Delta D_s \leq 1.50$
	6	$1.50 < \Delta D_s$
Population density changing		Take pairs of stores (j, j') opening in the same state, where $t_j \leq t_{j'} + 2$. Classify based on Popden_j (in units of 1,000 people within 5-mile radius). Define density classes 1, 2, 3, and 4 by $\text{Popden}_j < 15$, $15 \leq \text{Popden}_j < 40$, $40 \leq \text{Popden}_j < 100$, and $100 \leq \text{Popden}_j$.
	7	j in class 4, j' in class 3
	8	j in class 3, j' in class 2
	9	j in class 2, j' in class 1
	10	j in class 1, j' in class 2
	11	j in class 2, j' in class 3
	12	j in class 3, j' in class 4

^aNotes: The table uses the following notation: t_j is the opening date of store j , t_j^{state} is the opening date of the first store in the state where j is located, ΔD_s is the present value of the increment in distribution distance miles (in 1,000s) from doing the actual policy a^* instead of deviating and doing a . In words, to construct group 1, take the set of all stores opening when there is at least one store in their state that is 10 years old or more. For each such store, find alternative stores that open 3 or more years later in different states, where Wal-Mart has been in the different state no more than 4 years when the alternative store opens. Openings for general merchandise stores and food stores are considered two different opening events. In cases where a supercenter opens from scratch rather than as a conversion of an existing Wal-Mart, there are two opening events. In all the pairs considered, a general merchandise opening is paired with another general merchandise opening, and a food opening with another food opening.

TABLE X
SUMMARY STATISTICS OF DEVIATIONS BY DEVIATION GROUP

Deviation Group	Brief Description of Group	Number of Deviations	Mean Values			
			$\Delta \Pi$ (Millions of 2005 Dollars)	ΔD (Thousands of Miles)	ΔC_1 (log Popden)	ΔC_2 (log Popden ²)
	Store density decreasing					
1	$-.75 \leq \Delta D < 0$	64,920	-2.7	-.4	-.6	-3.0
2	$-1.50 \leq \Delta D < -.75$	61,898	-3.6	-1.1	-1.5	-9.0
3	$\Delta D < -1.50$	114,588	-4.7	-3.0	-3.4	-22.2
	Store density increasing					
4	$0 < \Delta D \leq .75$	158,208	3.0	.3	-1.9	-17.2
5	$.75 < \Delta D \leq 1.50$	34,153	3.7	1.0	-3.6	-28.9
6	$1.50 < \Delta D$	16,180	5.9	2.1	-4.8	-37.7
	Population density changing					
7	Class 4 to class 3	7,048	1.2	.0	3.2	31.1
8	Class 3 to class 2	10,435	3.7	.0	3.4	25.7
9	Class 2 to class 1	14,399	5.3	-.1	3.5	19.3
10	Class 1 to class 2	12,053	-2.4	.0	-3.4	-19.3
11	Class 2 to class 3	14,208	.6	-.1	-3.9	-29.4
12	Class 3 to class 4	14,877	2.5	.0	-4.6	-44.9
All	Weighted mean	522,967	-2	-.6	-2.1	-15.6

Read:

- Table IX defines the 12 deviation groups used for identification.
- Table X summarizes them. For example, in the store-density-decreasing groups, Wal-Mart gives up operating profit on average in exchange for lower distribution miles under the actual policy.
- These tables are important because they show the empirical content of the revealed-preference approach: identification is coming from many structured perturbations to the observed rollout.

5.7 Table XI: baseline estimates of density economies

Read:

- With only the 12 basic moments, the estimated lower and upper bounds on τ are **3.33** and **4.92**.
- Adding level 1 interactions narrows the bounds to **3.41–4.35**.
- Adding both level 1 and level 2 interactions narrows them further to the very tight baseline range of **3.50–3.67**.
- The corresponding 95% outer confidence thresholds are **2.97** and **5.04**.
- This is the main quantitative result of the paper.

TABLE XI
BASELINE ESTIMATED BOUNDS ON DISTRIBUTION COST τ^a

	Specification 1 Basic Moments (12 Inequalities)		Specification 2 Basic and Level 1 (84 Inequalities)		Specification 3 Basic and Levels 1, 2 (336 Inequalities)	
	Lower	Upper	Lower	Upper	Lower	Upper
Point estimate	3.33	4.92	3.41	4.35	3.50	3.67
Confidence thresholds						
With stage 1 error correction						
PPHI inner (95%)	2.69	6.37	2.89	5.40	3.01	4.72
PPHI outer (95%)	2.69	6.41	2.86	5.45	2.97	5.04
No stage 1 correction						
PPHI inner (95%)	2.84	5.74	2.94	5.11	3.00	4.62
PPHI outer (95%)	2.84	5.77	2.93	5.13	2.99	4.97

^aUnits are in thousands of 2005 dollars per mile year; number of deviations $M = 522,967$; number of store locations $N = 3,176$.

TABLE XII
MEAN INCREMENTAL MILES SAVED AND STORES SERVED FOR DISTRIBUTION CENTERS
ACROSS ALTERNATIVE OPENING DATES INCLUDING ACTUAL

	1 Year Prior to Actual	Actual Year Opened	1 Year After Actual	2 Years After Actual
All distribution centers ($N = 78$)				
Mean incremental miles saved	4.4	5.8	6.7	7.1
Mean stores served	23.6	52.1	58.4	62.9
By type of DC				
Regional distribution centers ($N = 43$)				
Mean incremental miles saved	6.1	7.7	8.7	8.9
Mean stores served	37.1	68.6	76.1	79.0
Food distribution centers ($N = 35$)				
Mean incremental miles saved	2.3	3.4	4.3	5.0
Mean stores served	6.9	31.8	36.5	43.0

TABLE XIII
ROBUSTNESS ANALYSIS OF ESTIMATED BOUNDS ON DISTRIBUTION COST τ^a

Specification	Number of Sample Deviations	Lower Bound		Upper Bound	
		Point Estimate	PPHI Outer Thresh.	Point Estimate	PPHI Outer Thresh.
Baseline	522,967	3.50	(2.97)	3.67	(5.04)
Split sample by opening date					
1962–1989	331,847	3.51	(2.92)	4.81	(7.19)
1990–2005	191,120	3.43	(1.85)	3.43	(3.90)
Vary operating profit parameters from first stage: net margin (baseline = .17)					
.15	522,967	2.62	(2.18)	3.05	(3.90)
.19	522,967	4.36	(3.78)	4.36	(6.22)
Labor cost factor					
20% higher	522,967	4.12	(3.64)	4.12	(5.66)
20% lower	522,967	2.80	(2.29)	3.38	(4.38)
Rent cost factor					
20% higher	522,967	3.60	(3.04)	3.79	(5.12)
20% lower	522,967	3.41	(2.89)	3.55	(5.03)
Exclude any deviations with change in supercenters	366,412	3.78	(2.66)	3.78	(5.61)
Include only deviations with change in supercenters	156,555	3.79	(1.52)	3.79	(4.72)
By supercenter opening					
1988–1997	59,693	.70	(.00)	.70	(3.48)
1998–2005	96,862	4.38	(2.48)	4.38	(5.39)

^aUnits are in dollars per mile year; all specifications include basic, level 1, and level 2 interactions (336 moment inequalities).

5.8 Table XII: distribution-center opening as a corroboration exercise

Read:

- At the actual opening date of a distribution center, the mean incremental store miles saved are about **5.8 thousand miles**, and the mean number of stores served is about **52.1**.
- Using a rough annual fixed-cost estimate of **\$18 million** for a distribution center implies $\hat{\tau} \approx \mathbf{3.10}$.
- This is close to the baseline estimate from the moment-inequality procedure, which is a strong corroboration because it comes from a different margin of choice.

5.9 Table XIII: robustness of the estimated bounds

Read:

- The baseline estimate remains stable when the sample is split into **1962–1989** and **1990–2005**; the estimates remain in the same broad range.
- Lowering the net margin from **0.17** to **0.15** reduces the point estimate to roughly **2.62–3.05**,

while increasing it to **0.19** raises it to about **4.36**.

- Varying labor costs by $\pm 20\%$ shifts the estimate, but the implied τ remains economically large.
- Excluding or isolating deviations involving supercenters gives estimates around **3.78–3.79**, except in the early 1988–1997 grocery period where the estimate falls to **0.70** with a very wide confidence interval, consistent with measurement-error concerns during Wal-Mart’s early food-distribution buildout.

6 Short summary

- **Method.** Estimate store-level demand and sales cannibalization, then use moment inequalities from pairwise deviations in the observed rollout to bound the value of distribution-related density economies.
- **Identification.** Wal-Mart’s actual store-opening pattern must weakly dominate alternative opening sequences once one accounts for incremental profits, store-to-distribution-center distance, and urban fixed-cost differences.
- **Application.** Wal-Mart store and distribution-center expansion in the United States, 1962–2005.
- **Main result.** The annual benefit of a store being 1 mile closer to its distribution center is tightly estimated at about **\$3.50–\$3.67 thousand per store-mile-year**, far larger than trucking costs alone and consistent with Wal-Mart’s dense contiguous rollout strategy.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that Wal-Mart’s observed inside-out expansion pattern is rationalized by **large economies of density**.
- It demonstrates that one can recover informative bounds on the value of those density economies even without observing the firm’s logistics costs directly.
- The moment-inequality revealed-preference approach is practical in a setting where a full dynamic solution would be computationally prohibitive.

7.2 Economic intuition

- Opening a new store close to existing stores is costly because it steals sales from those stores.
- But that same clustering lets Wal-Mart keep stores close to distribution centers and therefore makes the logistics network much cheaper and more responsive.
- Wal-Mart’s rollout reveals that the distribution-side benefit was strong enough to justify the demand-side cannibalization cost.

7.3 Takeaway

This paper is a clean example of **revealed-preference structural IO**. The big lesson is that the geometry of a firm's network can encode deep information about its cost structure. For Wal-Mart, dense expansion was not merely a by-product of growth. It was itself a major source of efficiency, and the value of that density was large enough to shape the entire national rollout path.